

Mobile App Policy Management

Create and execute customized policies to protect the enterprise from risky app behaviors and adhere to corporate policies.

App Behaviors List

GROUP	BEHAVIOR	ANDROID	IOS
Accesses Communication Channels	Accesses Internet - The application creates ports to establish network connections.	✓	✓
	Allows Sending Email - The application allows user to send an email from within the app, if user's device is already configured for sending email.		✓
	Sends SMS Messages - The application is capable of sending SMS message. This behavior is a common trait of malware applications.	✓	✓
	Includes 'Share via..' native controls - The app uses the interface provided by the OS to share files via different channels (e.g. Facebook, etc.)	✓	✓
	Makes failed network connections - The app makes network connections resulting in failed attempts (e.g. invalid IP address).	✓	✓
Accesses Sensitive Data	Accesses Address Book - The application accesses the address book. Improper handling of sensitive data by the application may disclose it to unintended parties.	✓	✓
	Accesses Calendar - The application accesses the device calendar. Improper handling of sensitive data by the application may disclose it to unintended parties.	✓	✓
	Accesses Call Logs - The application accesses to call logs on the device. Improper handling of sensitive data by the application may disclose it to unintended parties and compromise user's privacy.	✓	
	Accesses SMS Archive - The application read the SMS message body. Improper handling of sensitive data by the application may disclose it to unintended parties and compromise user's privacy.	✓	
	Accesses Telephony Service - The application accesses to Telephony Manager APIs to get information like device type, SMSC or country. These attributes are used to learn more about users and if used improperly will compromise user's privacy.	✓	
	Accesses UDID - The application accesses the Unique Device Identifier and International Mobile	✓	✓

GROUP	BEHAVIOR	ANDROID	IOS
	Equipment Identifier. These attributes are used to uniquely identify users and if used improperly will compromise user's privacy.		
	Accesses WiFi Info - The application requests access to ID of currently connected network.	✓	
	Gets Platform Version - The application gets access to mobile platform build and version information. A malware may use this information before attacking the device.	✓	
	Gets Recent Tasks - The application looks up the last applications the user ran on the mobile device. Improper use of this information, will compromise user's privacy.	✓	
	Reads Installed App Names - The application obtains the list of applications installed on the device. Improper use of this information, will compromise user's privacy.	✓	
	Tracks Location - The application accesses user's location. Improper handling of sensitive data by the application may disclose it to unintended parties.	✓	✓
	Accesses MDM configuration data - iOS provides a way for MDM to push configuration data to specific apps. This behavior detects apps that access this configuration data.		✓
	Reads data from the Clipboard - The app reads data available in the Clipboard before the app is launched.	✓	✓
Accesses Sensitive Data, Accesses Sensors	Accesses Bluetooth Sharing - The application uses Bluetooth to connect to Bluetooth peripheral devices.	✓	✓
	Accesses Camera - The application accesses the device camera. Improper handling of pictures by the application may disclose them to unintended parties.	✓	✓
	Accesses Microphone - The application accesses the device microphone. Improper handling of recordings by the application may disclose them to unintended parties.	✓	✓
Apple Entitlements	Has Embedded Provisioning Profile - The application includes a mobile provisioning profile. The existence of this file means the app was built for an ad-hoc or in-house (enterprise) distribution.		✓
	Supports Bonjour - The application includes support for service discovery using the Bonjour protocol.		✓
	Uses get-task-allow Flag - The application includes Get-Task-Allow entitlement. This flag		✓

GROUP	BEHAVIOR	ANDROID	IOS
	allows for an app to obtain access to resources belonging to other processes running on the same device.		
	Uses Keychain Access Groups - The application has access to keychain access groups. App developers use keychain access groups to share keychain data between their applications.		✓
Best Practice	Uses Common Crypto Libraries - The application uses CommonCrypto libraries that provide cryptographic capabilities to them.	✓	✓
	Uses Keychain - The application uses Apple Key Chain, which provides features to manage keys such as passwords.		✓
	Uses NIST NTP Servers - The application uses NIST recommended timeservers for setting its internal time.	✓	✓
	Requires Login - The app requires the user to enter credentials on a log in page to grant access.	✓	✓
	Uses SSL certificate pinning - The app uses SSL certificate pinning, which is considered best practice.	✓	✓
Best Practices (Continued)	Uses Security Frameworks - The application uses Apple Security Framework, which provides features for secure encryption of data, keychain access, and other cryptographic activities.		✓
Changes Device Configuration	Contains WatchKit apps/extensions - The app contains WatchKit apps/extensions.		✓
	Keyboard Extension with OpenAccess - Apps that contain keyboard extensions which require OpenAccess permission (access to storage, network, etc.)		✓
	Keyboard extensions - Apps that contain keyboard extensions and apps that are keyboard extensions	✓	✓
	Prevents device from locking screen - The app requests the device to not lock the screen. If the device falls into an attackers hands while on this state, access will be open to all data.	✓	✓
Code Vulnerability	Can Open other Applications - The application can handle a resource such as email or an image file by opening the corresponding application to open the resource and display it.		✓
	Discloses File Path to Source Code - The application includes file path to source code files. An adversary may use this information to learn about the internal structure of the app development organization's network.		✓

GROUP	BEHAVIOR	ANDROID	IOS
	Known Exploitable Application - The application is known to have a vulnerability that can get exploited and turn it into a malware.	✓	✓
	Loads Dynamic Code - The application might get code from external resources and run them dynamically. An adversary can exploit this vulnerability to run his own code on the device.	✓	
	Loads Native Libraries - The application may include native libraries to access and use resources outside of the Android Framework. This can be used to hide functionalities that are restricted by enterprise policies.	✓	
	Not Compiled as PIE - The application was not compiled as Position Independent Executable and it could expose them to memory corruption attacks.		✓
	References Private IPs - The application references an IP address in the private network range. An adversary may use these addresses to learn about the internal structure of the app development organization's network.		✓
	Replaces Random Number Generator - This application replaces the default seed for secure random number generator, which may cause it to return a predictable sequence of numbers, unfit for secure use.	✓	
	Stores credentials unencrypted - This app stores user credentials on device in clear text	✓	✓
	Supports opening document types - The app can open specific file types (e.g. PDF).	✓	✓
	Performs direct system calls - System calls are performed directly, instead of using standard and safer library calls.	✓	✓
	Performs dynamic symbol lookups - The app has dynamic links to load and lookup symbols and could potentially use private symbols. This technique can be used to prevent detection of unsecure functionality during the app vetting process.	✓	✓
	Uses JS Patch for hot patching - This application incorporates the JSPatch framework for hot patching. This allows app functionality to be updated without going through the iTunes vetting process.		✓
Code Vulnerability, Malware Indicator	Calls Exec When Running - The application executes commands outside the application sandbox. This behavior is a common trait in malware applications.	✓	

GROUP	BEHAVIOR	ANDROID	IOS
	Calls Exec When Running with Privilege - The application executes root commands outside the application sandbox. This behavior is a common trait in malware applications.	✓	
Commercial Transactions	Enables In App Purchases - The application support in app purchases which can lead to unintended charges.	✓	✓
Insecure Data Storage	Accesses External Storage - The application writes data to external storage media. An adversary can get access to data if external storage doesn't properly encrypt the data.	✓	✓
	Shares Files with iTunes - The application shares files with the user's computer through iTunes. An adversary can get access to data exploiting vulnerability on user's computer such as improper encryption.		✓
	Stores Data in Database on Device - The application uses a local database to store data on the device. An adversary can get access to data exploiting a vulnerability on user's device such as improper encryption.		✓
	Uses Hardcoded Credentials - The application includes a URL with a hardcoded username and password. An adversary can use the credentials to exploit the app or steal user's private data.	✓	✓
	HospitalGown: Redis - The application can communicate directly with a Redis server. Any data transmitted to or stored on the Redis server may be intercepted, copied, modified, or ransomed.		
	HospitalGown: ElasticSearch - The application can communicate directly with an ElasticSearch server. Any data transmitted to or stored on the ElasticSearch server may be intercepted, copied, modified, or ransomed.		
Malware Indicator	Changes Device Configuration - The application makes change to device bookmarks or desktop shortcuts. This behavior is a common trait in malware applications.	✓	
	Creates and Advertises a Service - The application can register a network service, such as printer or file server so that it can be found by name or service type or domain.		✓
	Creates Listening Ports - The application creates ports to listen to incoming connections. This behavior might allow an adversary to exploit a possible vulnerability.		✓
	Has Permission to Bind Device Admin - The	✓	

GROUP	BEHAVIOR	ANDROID	IOS
	application asks for device administration permission. This permission allows applications to build security aware enterprise applications.		
	Has Permission to Place Calls - The application asks for permission to place phone calls. A malware may use this permission to place unintended calls.	✓	
	Has Permission to Record Calls - The application asks for permission to record calls. A malware may use this permission to record calls and share with unintended parties.	✓	
	Has Permission to Send SMS - The application asks for permission to send or receive SMS. A malware may use this permission to send unwanted text messages.	✓	
	Has Permission to SignatureOrSystem - The application asks for SignatureOrSystem. Applications built into a system image use this permission to share specific features explicitly because they are built together.	✓	
Malware Indicator (Continued)	Has Permission to VPN Service - The application asks for permission to extend or build VPN solution. Malware may use this permission to intercept all data packets or to send data to unintended parties.	✓	
	Uses Aggressive Adware - This app is using aggressive Adware	✓	
	Writes the APN Settings - This permission allows applications to write the APN settings and it is not for use by third-party applications.	✓	
	Executes in the background - This application continues to execute in the background even when it has lost window focus.	✓	
	Handles Custom URL Schemes - This application has the ability to open custom URL schemes.	✓	✓
Malware/Spyware	Attempts Root Device - These malware applications attempt to run as superuser by exploiting a vulnerability in the operating system		✓
	Is a Spyware - The application silently monitors the activity and private data on the device. Examples of private data are email, recorded video and audio and contacts	✓	
	Malware Detected - The application performs malicious activity on the device.	✓	✓
	Infected by XcodeGhost - The application is affected by XcodeGhost malware		✓
	Uses mobiSage SDK with Backdoor - The application incorporates a version of the		✓

GROUP	BEHAVIOR	ANDROID	IOS
	mobiSage SDK that is known to collect private user's information (iBackDoor)		
	Infected by AceDeceiver – The application is affected by AceDeceiver malware that targets non-jailbroken devices, enables spear phishing attacks and allows attackers to install trojan apps on device		✓
	Infected by Dresscode – The application creates a botnet that is used to disguise ad clicks and generate false traffic, generating revenue for the attacker	✓	
Sends data to third-party services	Cloud Storage Default - The application stores files in a cloud. Files are sent over the network to third-party cloud file storage.		✓
	Connects to Facebook Social Network - The application can share sensitive information on Facebook and can obtain a user private data from Facebook.	✓	✓
	Connects to Google Social Network - The application can share sensitive information on Google Plus and can obtain a user private data from Google Plus	✓	✓
Sends Data to third-party Services (Continued)	Connects to Social Networking Platforms - The application can share sensitive information on Social Networking Platforms and can obtain a user private data from them.	✓	✓
	Connects to Twitter Social Network - The application can share sensitive information on Twitter and can obtain a user private data from Twitter.	✓	✓
	Connects to Weibo Social Network - The application can share sensitive information on Weibo and can obtain a user private data from Weibo.	✓	✓
	Displays Ads - The application uses third party frameworks to display advertisement. This application may share sensitive data with Ad providers.	✓	✓
	Is a Cloud Storage Provider - The application is cloud storage providers.		✓
	Sends Analytics Data to Flurry - The application collects and reports usage information to Flurry servers. Examples of usage information are location of users and how apps are used.	✓	✓
	Sends Analytics Data to Google - The application collects and report usage information to Google servers. Examples of usage information are location of users and how apps are used.	✓	✓

GROUP	BEHAVIOR	ANDROID	IOS
	Sends Crash Report to Bugsense - The application shares its crash logs with Bugsense, a service for monitoring application health and availability. Crash logs can include private data.	✓	✓
	Sends Crash Report to Crashlytics - The application shares its crash logs with Crashlytics, a service for monitoring application health and availability. Crash logs can include private data.	✓	✓
	Sends Crash Report to Crittercism - The application shares its crash logs with Crittercism, a service for monitoring application health and availability. Crash logs can include private data.	✓	✓
	Sends Crash Report to Flurry - The application shares its crash logs with Flurry, a service for monitoring application health and availability. Crash logs can include private data.	✓	✓
	Sends Crash Report to Google - The application shares its crash logs with Google Analytics, a service for monitoring application health and availability. Crash logs can include private data.	✓	✓
	Sends data to Analytic Frameworks - The application collects and reports usage information to Analytics Frameworks. Examples of usage information are location of users and how apps are used.	✓	✓
Sends Data to third-party Services (Continued)	Sends data to Crash Reporting Frameworks - The application shares its crash logs with Crash Reporting Frameworks, for monitoring application health and availability. Crash logs can include private data.	✓	✓
	Stores Files in Box - The application uses Box to allow the user to transfer files to and from the device.	✓	✓
	Stores Files in Cloud - The application uses third party cloud services to allow the user to transfer files to and from the device.	✓	✓
	Stores Files in Dropbox - The application uses Dropbox to allow the user to transfer files to and from the device.	✓	✓
	Stores Files in Evernote - The application uses Evernote to allow the user to transfer files to and from the device.	✓	✓
	Stores Files in Google Drive - The application uses Google Drive to allow the user to transfer files to and from the device.	✓	✓
	Stores Files in iCloud - The application uses iCloud to allow the user to transfer files to and from the device.	✓	✓

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	Stores Files in Skydrive - The application uses SkyDrive to allow the user to transfer files to and from the device.	✓	✓
	Stores Files in Sugarsync - The application uses SugarSync to allow the user to transfer files to and from the device.	✓	✓
	Uses Admob Advertising Provider - The application uses Admob Advertising platform to display advertisement. Applications may share sensitive data with Ad providers.	✓	✓
	Uses Greystripe Advertising Provider - The application uses Greystripe Advertising platform to display advertisement. Applications may share sensitive data with Ad providers.	✓	✓
	Uses iAd Advertising Provider - The application uses iAd Advertising platform to display advertisement. Applications may share sensitive data with Ad providers.		✓
	Uses Mobclix Advertising Provider - The application uses Mobclix Advertising platform to display advertisement. Applications may share sensitive data with Ad providers.	✓	✓
	Connects to LinkedIn Social Network - The application can share sensitive information on LinkedIn and can obtain a user's private data from LinkedIn.		✓
Sends Data to third-party Services (Continued)	Uses YouMi SDK - The application incorporates YouMi Ad SDK. This application is collecting information using private iOS APIs.		✓
Suspicious Malware/Adware Indicator	Detects if device is rooted/jailbroken - The app checks if the device has already been rooted/jailbroken. This knowledge could be used by the app to execute privileged commands.	✓	✓
Unsafe Data Transmission	Sends Data Unencrypted - The application creates network connections that don't use SSL. Improper encryption of data during transfer or on end server may disclose them to unintended parties.	✓	
	Sends PII - The application transmits private data like IMEI/UDID, IP address or phone number to a server.	✓	✓
	Sends Sensitive Data without Encryption - The application transmits the sensitive data without proper encryption or as a query parameter. This vulnerability allows an adversary to eavesdrop the data.	✓	✓
	Disables the SSL CA validation - The app does not validate SSL certificate authorities, which can	✓	✓

GROUP	BEHAVIOR	ANDROID	IOS
	increase the risk of unsecure connections.		
	Does not enforce HTTPS connections - The app does not use secure HTTPS connections during communications.	✓	✓
Uploads Sensitive Data	Sends Address Book - The application sends sensitive data and PII from address book over the network. Improper encryption of data during transfer or on end server may disclose them to unintended parties.	✓	✓
	Sends Calendar - The application sends sensitive data and PII from calendar over the network. Improper encryption of data during transfer or on end server may disclose them to unintended parties.	✓	✓
	Sends IMEI - The application sends the device IMEI (International Mobile Equipment Identifier) over the network. These attributes are used to uniquely identify users and if used improperly will compromise user's privacy.	✓	
	Sends Location - The application sends the device location over the network. Improper encryption of data during transfer or on end server may disclose them to unintended parties.	✓	✓
	Sends credentials unencrypted - The app sent the user's provided credentials clear text over the network.	✓	✓
	Sends the device's name - The app often sends the name of the owner (e.g., John Smith's iPhone).	✓	✓
	Sends UDID - The application sends the device UDID (Unique Device Identifier) over the network. These attributes are used to uniquely identify users and if used improperly will compromise user's privacy.		✓
Uses third-party SDK	Uses Apperian SDK - The application incorporates the Apperian EASE SDK. This SDK is used to provide support of EASE platform inside iOS apps.		✓
	Uses EnsureIT SDK - The application uses EnsureIT protection features including obfuscation, checksum and anti-debugging functionality to strengthen and protect them from tampering.		✓
	Uses MobileIron SDK - The application incorporates MobileIron AppConnect SDK. This SDK allows for an app to be managed by the MobileIron Mobile Device Management platform	✓	✓
Uses third-party SDK	Uses Mocana SDK - The application is wrapped		✓

GROUP	BEHAVIOR	ANDROID	IOS
(Continued)	by Mocana's Mobile App Protection that provides secure file storage and network transmission, among other features		
	Uses MonoTouch SDK - The application incorporates the MonoTouch Framework. This framework is used to develop cross platform mobile apps.		✓
	Uses PhoneGap SDK - The application incorporates PhoneGap Mobile Framework. Third party frameworks may include vulnerabilities that can be exploited for malicious activities.	✓	✓
	Uses Titanium SDK - The application incorporates the Titanium Framework. This SDK is used to develop cross platform mobile apps.	✓	